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Choice of dispersion coefficients in reactive transport calculations on smoothed fields

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Abstract

Local dispersion dominates the mixing of compounds that are introduced separately into the subsurface and do not partition into any other than the aqueous phase. Thus, reactions between these compounds are controlled by dispersive mixing if they are not limited by kinetics. I quantify longitudinal dispersive mixing by the longitudinal effective dispersion coefficient of a conservative tracer introduced by a point-like injection [Water Resour. Res. 36 (12) (2000) 3591–3604]. In the upscaling of mixing-controlled reactive transport, I apply the mean velocity and the effective dispersion coefficient to the macroscopic transport calculations, whereas the reactive parameters on the macro-scale are identical to those on the local scale. The applicability of the approach is demonstrated for the transport of compounds undergoing a second-order irreversible bimolecular reaction. Ten realizations of a two-dimensional heterogeneous log-conductivity field are considered. Using the effective dispersion parameters, the overall mass balance is met in the ensemble average, whereas solute spreading is underestimated. I assess the lack of spreading by the difference between the expected macrodispersion and effective dispersion coefficients. I extend the approach to simulations on log-conductivity fields obtained by kriging of regularly spaced conductivity measurements. These fields contain the large-scale features of the true fields but do not resolve the small-scale variability. For the calculations on the kriged fields, the corresponding conditional covariance is substituted into the analytical expressions of effective dispersion, yielding a correction effective dispersion coefficient. The comparison between simulations on the fully resolved fields and on the kriged fields indicates that the approach is valid for wide plumes meeting the ergodicity condition. The high variability of mixing on small scales unresolved by kriging, however, leads to severe uncertainty when mixing-controlled reactions are predicted for narrow plumes.
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1. Introduction

The heterogeneity of natural formations determines the spreading of solutes in groundwater. In stationary media with a small variance of the log-conductivity fluctuations, we can predict macrodispersion, that is, the spreading of a large solute plume, quite accurately by linear stochastic theory (Gelhar and Axness, 1983; Neuman et al., 1987; Dagan, 1988). In recent years, these approaches have been refined to predict the effective dispersion of smaller plumes (Dagan, 1990; Rajaram and Gelhar, 1993; Attinger et al., 1999; Dentz et al., 2000a,b) and the concentration variance (Kapoor and Gelhar, 1994; Fiori and Dagan, 2000).

While the concept of macrodispersion has been shown appropriate in many conservative-tracer studies, it has led to erroneous results in the transport of multiple compounds reacting with each other (Molz and Widdowson, 1988). A basic requirement for reactions to take place is the mixing of the reacting compounds. The mixing must occur on the scale of reactions, that is, on the pore-scale. Macrodispersion, however, describes how fast the second central spatial moments of a very large plume increase. The latter quantity includes both mixing on the local scale and spreading on a larger scale, i.e., the increasing irregularity of the plume shape. If we apply the Fickian macrodispersion concept to reactive transport, we merge mixing and spreading, overestimate the rate of mixing, and therefore arrive at effective reaction rates which are too large (Molz and Widdowson, 1988; Ginn et al., 1995; Kapoor et al., 1997).

In many practical applications, mixing on the local scale is determined by differences in sorption behavior of the reacting compounds (see, e.g., the asymptotic analysis of Oya and Valocchi (1998) for stratified formations). In these cases, local-scale dispersion may be neglected and reactive transport can be predicted by the stochastic–convective model (Simmons et al., 1995), which has been applied to bioreactive model problems by Ginn et al. (1995), Xin and Zhang (1998), and Cirpka and Kitanidis (2001) among others. The framework is essentially identical to that of Dagan and Cvetkovic (1996). For the case of negligible local dispersion, stochastic-analytical results of bioreactive transport have also been gained by Eulerian analysis (Miralles-Wilhem et al., 1997; Miralles-Wilhem and Gelhar, 2000). If the interacting compounds do not undergo interphase mass transfer, however, local dispersion remains the only mixing process and analytical results neglecting it are not applicable. Ginn (2001) extended the stochastic–convective approach to include longitudinal pore-scale dispersion within noninteracting streamtubes. In dilution and mixing in heterogeneous domains, however, pore-scale dispersion into the transverse direction is more important than that into the longitudinal direction (Kitanidis, 1994; Cirpka and Kitanidis, 2000b). The variability of the flow field leads to extensive stretching of the plume interfaces (Urroz et al., 1995; Weeks and Sposito, 1998), so that transverse pore-scale dispersion acts over a much larger area than longitudinal dispersion. In a quasi one-dimensional set up, where an aqueous solution containing one reactant is replaced by a solution containing the other, pore-scale transverse dispersion transfers longitudinal spreading, that is, the stretching of the interface, to longitudinal mixing, i.e., the overlapping of the plumes. In the advective–dispersive streamtube approach of Cirpka and Kitanidis (2000a), the effect of pore-scale transverse dispersion on longitudinal mixing is quantified by the second central moment of conservative breakthrough curves

obtained from point measurement. The derived measure of longitudinal mixing was used as an apparent dispersivity of mixing in a parallel streamtube concept. The latter approach relies on point measurements of concentrations. In the present study, I apply recent results of stochastic linear theory (Dentz et al., 2000a; Fiori and Dagan, 2000) to predict mixing-related dispersion coefficients from the geostatistical characterization of the log-conductivity field.

Kapoor et al. (1997) analyzed bimolecular reactions of non-sorbing compounds in heterogeneous domains. They derived a macroscopic transport equation including an effective reaction coefficient depending on the cross-covariance between the concentrations of the reacting compounds. The cross-covariance of concentrations undergoes an additional transport equation that is similar to that of the single-compound concentration variance (Kapoor and Gelhar, 1994; Kapoor and Kitanidis, 1998). Unfortunately, the governing equation for the concentration cross-covariance included correlation terms that are difficult to determine. Kapoor and Anmala (1998) defined lower bounds on the decisive variance dissipation term. These theoretical studies agreed with experimental results on bimolecular reactions in capillary tubes (Kapoor et al., 1998) and in soil columns (Raje and Kapoor, 2000).

The previously mentioned macroscopic transport equations (Kapoor et al., 1997; Miralles-Wilhem et al., 1997; Miralles-Wilhelm and Gelhar, 2000) include a macrodispersion tensor, guaranteeing the correct second central moments, and an effective reaction coefficient, so that the overall reaction rate is correct even though averaged concentrations are used in their evaluation. In the following, I will use a different approach. I keep the original reaction terms and use a dispersion tensor that leads to the correct rate of mixing rather than second central moments. Then, the overall mass-balance is correct whereas the second central moments are underestimated. We can approximate the missing amount of spreading from the difference between the dispersion tensor used and the conservative macrodispersion tensor. The advantage of my approach is that I avoid the derivation of lengthy terms describing the effective reaction rates as function of ensemble-averaged concentrations. Such terms might become extremely complicated for complex reactions.

The dispersion coefficient for which I expect the correct amount of mixing is the effective dispersion coefficient for a point-like injection (Dentz et al., 2000a). This is illustrated schematically in Fig. 1. The concentration distributions of two compounds A and B are shown in the top half of Fig. 1. At the initial state, shown on the left-hand side, the two compounds are separated. To the left of $x=0$, we find only compound A, and to the right only compound B. These two plumes move within a heterogeneous flow field. The concentration distribution at time t is shown on the right-hand side of Fig. 1. Because of heterogeneity, the interface between the two plumes has become irregular in shape. The irregularity of the interface referred to as spreading, however, is not a measure of mixing. In mixing, the important question is whether, and to which extent, compounds A and B occur at the same point at the same time so that they may react with each other. For this purpose, we have to track back the concentrations observed at a point at time t to where it is coming from at the initial state. This is illustrated in the third line of Fig. 1. Due to local dispersion, and enhanced significantly by the variability of the flow field, the area influencing the point measurement increases when we go back to earlier times. In Fig.

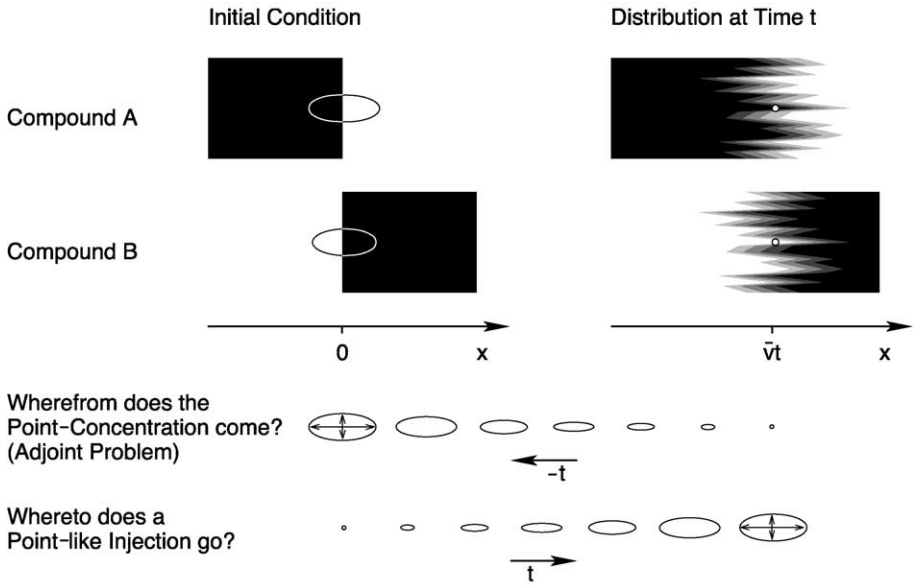

Fig. 1. Mixing of solutes and its approximation by effective dispersion for a point-like injection.

1, the latter is illustrated by an increasing ellipse. In reality, of course, the area of influence is neither elliptic in shape nor sharply bounded but is a smooth distribution with a decreasing rate of influence the further away we go from the center of the influencing area. Starting at the mean arrival time at the point observation and going back to time zero, the area of influence is half in the region containing only compound A and half in that containing compound B. The length of the ellipse quantifies the amount of mixing.

Determining the area influencing a point measurement is an adjoint problem. In the corresponding forward problem, illustrated in the bottom line of Fig. 1, we inject a solute by a point source at time zero and track it forward in time. At the mean travel time needed to reach the point of observation, the area influenced by the point injection has been extended. Half the rate of change of the expected longitudinal second central spatial moment is referred to as longitudinal effective dispersion coefficient D_ℓ^e (Dentz et al., 2000a). If we consider the multi-dimensional moments, we arrive at the effective dispersion tensor $\mathbf{D}^e$. Stochastic-analytical results for $\mathbf{D}^e$ have been developed recently by Dentz et al. (2000a) and Fiori and Dagan (2000). In stationary flow fields, the second central moment of a fictitious point-related plume is identical to that of the area of influence considered in the back-tracking problem described above. That is, we may use the stochastic-analytical results of effective dispersion for a point-like injection to approximate solute mixing by pore-scale dispersion in heterogeneous domains.

Cirpka and Nowak (in press) claimed—without further proof—that "in calculations of mixing-controlled reactive transport, $\mathbf{D}^e(t)$ is the macroscopic dispersion parameter that guarantees the accurate mass balance, although the spatial moments of the reactive plumes will be underestimated". The objective of the present study is to test this hypothesis by

numerical simulations. In particular, I want to test whether the approaches of Cirpka and Nowak (in press) for effective dispersion on smoothed log-hydraulic conductivity fields are applicable to mixing-controlled reactive transport simulations.

2. Theory

2.1. Dispersion coefficients

Consider the log-conductivity field $Y(\mathbf{x}) = \ln(K(\mathbf{x}))$ a stationary random function characterized by its mean $\bar{Y}$ and covariance $R_{YY}(\mathbf{h})$:

$$\bar{Y} = \langle Y \rangle \neq f(\mathbf{x}) \tag{1}$$

$$R_{YY}(\mathbf{h}) = \langle (Y(\mathbf{x}) - \bar{Y})(Y(\mathbf{x}+\mathbf{h}) - \bar{Y}) \rangle \neq f(\mathbf{x}) \tag{2}$$

in which $\langle\rangle$ denotes the expected-value operator, and $\mathbf{h}$ is the separation vector. The Fourier transform of the covariance function $R_{YY}(\mathbf{h})$ is the power spectral density function of the log-conductivity fluctuations $S_{YY}(\mathbf{s})$:

$$S_{YY}(\mathbf{s}) = \frac{1}{(2\pi)^{d/2}} \int_{V_\infty} R_{YY}(\mathbf{h}) \exp(i\mathbf{h}\cdot\mathbf{s}) d\mathbf{h} \tag{3}$$

in which $\mathbf{s}$ is the vector of wave numbers, d is the dimensionality of the domain, and i is the imaginary number. I assume a uniform gradient $-\mathbf{J} = \nabla\bar{\phi}$ of the mean head $\bar{\phi}$. Then, the spectrum $\mathbf{S_{vv}}(\mathbf{s})$ of all seepage-velocity components v_i can be evaluated from that of log-conductivity $S_{YY}(\mathbf{s})$ by (Bakr et al., 1978; Gelhar and Axness, 1983):

$$\mathbf{S_{vv}}(\mathbf{s}) = \left(\mathbf{J} - \frac{\mathbf{s}\cdot\mathbf{J}}{s^2}\mathbf{s}\right)\left(\mathbf{J} - \frac{\mathbf{s}\cdot\mathbf{J}}{s^2}\mathbf{s}\right)^T \frac{K_g^2}{\theta^2} S_{YY}(\mathbf{s}) \tag{4}$$

in which $K_g = \exp(\bar{Y})$ is the geometric mean of the hydraulic conductivity K, θ is the volumetric water content, and $s = \sqrt{\mathbf{s}\cdot\mathbf{s}}$.

The macrodispersion tensor $\mathbf{D}^*(t)$ is defined as the dispersion tensor needed to match the second central spatial moments of a very large plume in a heterogeneous domain when transport is described by the uniform advection–dispersion equation. In stationary velocity fields with small log-conductivity fluctuations, linear stochastic theory yields (Gelhar and Axness, 1983; Neuman et al., 1987; Dagan, 1988):

$$\mathbf{D}^*(t) = \mathbf{D} + \frac{1}{(2\pi)^{d/2}} \int_{V_\infty} (\mathbf{s}^T\mathbf{D}\mathbf{s} - i\mathbf{s}\cdot\bar{\mathbf{v}}) \frac{1 - \exp(-t(\mathbf{s}^T\mathbf{D}\mathbf{s} + i\mathbf{s}\cdot\bar{\mathbf{v}}))}{(\mathbf{s}\cdot\bar{\mathbf{v}})^2 + (\mathbf{s}^T\mathbf{D}\mathbf{s})^2} \mathbf{S_{vv}}(\mathbf{s}) d\mathbf{s} \tag{5}$$

in which $\mathbf{D}$ is the local dispersion tensor assumed uniform, $\bar{\mathbf{v}}$ is the mean seepage velocity, and t is time.

For plumes of initially small size, the macrodispersion tensor $\mathbf{D}^*(t)$ applies to the moments of ensemble-averaged concentrations. That is, we consider the ensemble of an infinite number of aquifer realizations sharing the same geostatistical description. In a thought experiment, we calculate flow and transport in all realizations and average the concentrations over all possible realizations prior to analyzing the spatial moments of the plume. The order of averaging and calculating the moments is reversed in the analysis of effective dispersion (Attinger et al., 1999; Dentz et al., 2000a,b). For a plume of initial size $\mathbf{V}_0$, we first take the second central moments and then ensemble-average half the rate of change in order to evaluate the effective dispersion tensor $\mathbf{D}^e(t,\mathbf{V}_0)$, which is a function of time and the initial volume of the plume $\mathbf{V}_0$. In the following, I consider only the case of a point-like injection omitting $\mathbf{V}_0$ in the notation.

Without local dispersion, $\mathbf{D}^e(t)$ would remain zero at all times. Accounting for local dispersion, the effective dispersion tensor is initially identical to the local dispersion tensor, grows over time since local transverse dispersion makes the plume sample increasingly more streamlines, and finally reaches the asymptotic macrodispersion tensor, although at very large times. Dentz et al. (2000a) derived the following analytical expression of $\mathbf{D}^e(t)$ in an Eulerian, and Fiori and Dagan (2000) in a Lagrangian framework:

$$\mathbf{D}^*(t) - \mathbf{D}^e(t) = \frac{1}{(2\pi)^{d/2}} \int_{V_\infty} \frac{(\mathbf{s}^T\mathbf{D}\mathbf{s} - i\mathbf{s}\cdot\overline{\mathbf{v}})(\mathbf{exp}(t(i\mathbf{s}\cdot\overline{\mathbf{v}} - \mathbf{s}^T\mathbf{D}\mathbf{s})) - \mathbf{exp}(-2t\mathbf{s}^T\mathbf{D}\mathbf{s}))}{(\mathbf{s}\cdot\overline{\mathbf{v}})^2 + (\mathbf{s}^T\mathbf{D}\mathbf{s})^2} \mathbf{S}_{\mathbf{vv}}(\mathbf{s})d\mathbf{s} \tag{6}$$

For the isotropic Gaussian covariance model of log-conductivity with correlation length λ and a small scalar local dispersion coefficient D, Dentz et al. (2000a) gave the following approximation for the longitudinal effective dispersion coefficient:

$$D_\ell^e(t) = D + (D_\ell^*(\infty) - D)\left(1 - \left(\frac{\lambda^2}{\lambda^2 + 4tD}\right)^{(d-1)/2}\right) \tag{7}$$

in which $D_\ell^*(\infty)$ is the asymptotic longitudinal macrodispersion coefficient.

The effective dispersion tensor $\mathbf{D}^e$ defined above is an ensemble-averaged quantity. In single realizations, a point source will spread differently if we introduce it at different points. If the integral scale of the flow field is much smaller than the width of the plume, ensemble averages may be replaced by cross-sectional averages (ergodicity condition). Taking the longitudinal effective dispersion coefficient D_ℓ^e resulting from stochastic theory as a measure of mixing, we can therefore estimate only the mean longitudinal mixing behavior of a plume wide enough to meet the ergodicity condition.

2.2. Conditioning

The macrodispersion and effective dispersion tensors, $\mathbf{D}^*(t)$ and $\mathbf{D}^e(t)$, derived by Eqs. (3)–(6) give the expected amount of plume spreading and mixing in aquifers that are described exclusively by the geostatistical parameters of the log-conductivity distribution.

In order to evaluate the mean $\bar{Y}$ and the covariance $R_{YY}(\mathbf{h})$ of the log-conductivity, we need measurements of Y or correlated quantities. In the following transport calculations, however, the actual measured values are neglected altogether. That is, we apply $\mathbf{D}^*(t)$ and $\mathbf{D}^e(t)$, derived by the equations given above, to transport calculations in which a uniform log-conductivity value of $\bar{Y}$ is used. In this context, the covariance $R_{YY}(\mathbf{h})$ describes the uncertainty associated with using $\bar{Y}$ rather than the true values $Y(\mathbf{x})$.

Actual measurements $\hat{\mathbf{Y}}$ of $Y(\mathbf{x})$ give us more detailed information than the geostatistical parameters. We may use them to condition the generation of random auto-correlated log-conductivity fields used in Monte-Carlo simulations, or we may interpolate the measured values to get the best linear unbiased estimate of $Y(\mathbf{x})$ meeting local measurements $\hat{\mathbf{Y}}$. The latter procedure is well known as kriging. In both, the generation of conditional realizations and kriging, the geostatistical parameters can be inferred from the data by statistical cross-evaluation techniques such as the method of orthonormal residuals (Kitanidis, 1991).

Since Monte-Carlo simulations are computationally costly, I will perform reactive transport calculations on kriged log-conductivity fields. On such fields, the spatial variability of $Y(\mathbf{x})$ is partially recovered, and the uncertainity in Y smaller than expressed by $R_{YY}(\mathbf{h})$. Therefore, I parameterize only the effect of the unresolved variability on solute spreading and dispersive mixing, whereas the macrodispersion and effective dispersion coefficients derived from $R_{YY}(\mathbf{h})$ overestimate the second central moments of solute plumes on the interpolated field.

For interpolation, I apply the function estimate form of kriging using the following interpolator (see, e.g., Kitanidis, 1996):

$$\tilde{\mathbf{Y}}(\mathbf{x}) = \beta + \mathbf{R}_{Y(\mathbf{x})\hat{\mathbf{Y}}}\boldsymbol{\xi} \tag{8}$$

in which $\tilde{\mathbf{Y}}(\mathbf{x})$ is the estimated log-conductivity value at location $\mathbf{x}$, β is the estimated mean log-conductivity, $\mathbf{R}_{Y(\mathbf{x})\hat{\mathbf{Y}}}$ is a row vector of the covariance between Y at $\mathbf{x}$ and at all measurement locations $\hat{\mathbf{x}}$, and $\boldsymbol{\xi}$ is a column vector of weights associated with the individual measurements. $\boldsymbol{\xi}$ and β are the solution of the kriging system of equations:

$$\begin{bmatrix} \mathbf{R}_{\hat{\mathbf{Y}}\hat{\mathbf{Y}}} & \mathbf{u} \\ \mathbf{u}^T & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{\xi} \\ \beta \end{bmatrix} = \begin{bmatrix} \hat{\mathbf{Y}} \\ 0 \end{bmatrix} \tag{9}$$

in which $\mathbf{u}$ is a vector consisting only of unity entries, $\mathbf{R}_{\hat{\mathbf{Y}}\hat{\mathbf{Y}}}$ is the covariance matrix between all log-conductivity measurements, and $\hat{\mathbf{Y}}$ is the vector of measured values. The uncertainty associated with using the best estimate $\tilde{\mathbf{Y}}(\mathbf{x})$ rather than the true value $Y(\mathbf{x})$ is given by the conditional covariance $R_c(\mathbf{x}_k,\mathbf{x}_\ell)$ calculated by (Kitanidis, 1996):

$$R_c(\mathbf{x}_k, \mathbf{x}_\ell) = R_{YY}(\mathbf{x}_k - \mathbf{x}_\ell) - [\mathbf{R}_{Y(\boldsymbol{x}\boldsymbol{k})_{\hat{\mathbf{Y}}}}, 1] \begin{bmatrix} \mathbf{R}_{\hat{\mathbf{Y}}\hat{\mathbf{Y}}} & \mathbf{u} \\ \mathbf{u}^T & 0 \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{R}^T_{Y(\boldsymbol{x}_\ell)\hat{\mathbf{Y}}} \\ 1 \end{bmatrix} \tag{10}$$

The conditional covariance $R_c(\mathbf{x}_k,\mathbf{x}_\ell)$ is nonstationary, that is, it depends on the actual positions $\mathbf{x}_k$ and $\mathbf{x}_\ell$ rather than the separation vector $\mathbf{x}_k - \mathbf{x}_\ell$. For a rather dense regular grid

of measurement points, however, we may remove the nonstationarity of $R_c(\mathbf{x}_k,\mathbf{x}_\ell)$ by taking spatial averages (Cirpka and Nowak, in press):

$$\bar{R}_c(\mathbf{h}) = \lim_{L_1,L_2,L_3\to\infty} \frac{1}{L_1L_2L_3}\int_{-L_1/2}^{+L_1/2}\int_{-L_2/2}^{+L_2/2}\int_{-L_3/2}^{+L_3/2} R_c(\mathbf{x},\mathbf{x}+\mathbf{h})dx_3dx_2dx_1 \tag{11}$$

Cirpka and Nowak (in press) outlined an efficient method for the calculation of $\bar{R}_c(\mathbf{h})$ in periodic domains with regular spacing of measurement points that will be used in the following application. $\bar{R}_c(\mathbf{h})$ quantifies, in an averaged form, the spatial variability that has been removed by kriging. The variability recovered on the kriged field is $R_{YY}(\mathbf{h}) - \bar{R}_c(\mathbf{h})$. When we perform transport simulations on kriged log-conductivity fields, the latter variability causes spreading of the solute plume and a related increase of solute mixing, but both mixing and spreading are smaller than on the fully resolved log-conductivity field.

The expressions for macrodispersion, Eq. (5), and effective dispersion, Eq. (6), depend via Eqs. (3) and (4) linearly on the covariance of log-conductivity $R_{YY}(\mathbf{h})$. As a consequence, we can split $R_{YY}(\mathbf{h})$ into any sum of components, evaluate the effect of each component on $\mathbf{D}^*(t)$ and $\mathbf{D}^e(t)$ separately by Eqs. (3)–(6), and finally sum up all contributions to dispersion.

When we use the local dispersion tensor $\mathbf{D}$ in transport calculations on kriged hydraulic conductivity fields, we expect that half the rate of change of the second central spatial moments meets the macrodispersion and effective dispersion tensors as calculated from substituting $R_{YY}(\mathbf{h}) - \bar{R}_c(\mathbf{h})$ into Eqs. (3)–(6). I refer to the missing amount of dispersion as the correction macrodispersion tensor $\mathbf{D}_c^*(t)$ and the correction effective dispersion tensor $\mathbf{D}_c^e(t)$, respectively (Cirpka and Nowak, in press). Its values are calculated by substituting $\bar{R}_c(\mathbf{h})$ into Eqs. (3)–(6) and subtracting the local dispersion tensor $\mathbf{D}$. Table 1 summarizes the definitions of the different dispersion tensors. Cirpka and Nowak (in press) suggested to use $\mathbf{D}+\mathbf{D}_c^e(t)$ in transport calculations on kriged conductivity fields in order to retrieve the correct amount of mixing. They showed the validity of the approach by comparing two-particle moments evaluated numerically on fully resolved fields with those on kriged fields applying the correction effective dispersion tensor. In the current study, I apply the approach to reactive transport.

Table 1
Definition and application of dispersion tensors

$\mathbf{D}$	local dispersion tensor		apply on highly resolved fields
$\mathbf{D}^*$	macrodispersion tensor	expected half rate of change of second central spatial moments of a very large plume	apply on equivalent uniform fields to meet solute spreading
$\mathbf{D}^e$	effective dispersion tensor	expected half rate of change of second central spatial moments of a plume originating from a point-like injection	apply on equivalent uniform fields to meet solute mixing
$\mathbf{D}_c^*$	correction macrodispersion tensor	missing macrodispersion when applying $\mathbf{D}$ on interpolated fields	add to $\mathbf{D}$ to meet solute spreading on interpolated fields
$\mathbf{D}_c^e$	correction effective dispersion tensor	missing effective dispersion when applying $\mathbf{D}$ on interpolated fields	add to $\mathbf{D}$ to meet solute mixing on interpolated fields

In practical applications, in which the approach by Cirpka and Nowak (in press) to calculate $\bar{R}_c(\mathbf{h})$ may not be suitable, one can determine $R_{YY}(\mathbf{h}) - \bar{R}_c(\mathbf{h})$ by analyzing the interpolated log-conductivity field $\tilde{Y}(\mathbf{x})$ using standard geostatistical methods. Then, we treat the interpolated field $\tilde{Y}(\mathbf{x})$ as a stationary one and determine the auto-covariance among all interpolated values as function of the separation vector $\mathbf{h}$. Since the number of discretized interpolated values $\tilde{Y}(\mathbf{x})$ is much higher than the number of measurements $\hat{\mathbf{Y}}$, it may be advisable to evaluate directly the spectrum $S_{\tilde{Y}\tilde{Y}}(\mathbf{s})$ of the interpolated field by Fourier techniques known from image processing (Brigham, 1974). The spectrum corresponding to the averaged conditional covariance function $\bar{R}_c(\mathbf{h})$ is then given by subtracting $S_{\tilde{Y}\tilde{Y}}(\mathbf{s})$ from the Fourier transform $S_{YY}(\mathbf{s})$ of the unconditional covariance function $R_{YY}(\mathbf{h})$.

3. Application to reactive transport

3.1. Model domain

As an approximation to stationary media, I use periodic domains (Kitanidis, 1992). Both stationary and periodic fields can be statistically characterized by their auto-covariance function $R_{YY}(\mathbf{h})$. Although both describe an infinite domain, only the periodic field can be completely represented through a single unit cell, which has finite dimensions. In fact, a stationary field can be viewed as a periodic one at the limit of a unit cell with infinite dimensions. Cirpka and Nowak (in press) generated 50 periodic fields of 1000×500 cells by a spectral approach similar to that of Dykaar and Kitanidis (1992). The parameters of the fields are given in Table 2. Since reactive-transport calculations are computationally more costly than conservative-transport simulations, I use only the first 10 realizations for reactive transport. Fig. 3 shows a clip of four unit cells of a particular realization of the periodic domain. For each realization, I "take measurements" by taking the true values of the fields at points on a regular grid with a spacing of $\Delta x_{\mathrm{obs}} = 1.25\lambda$ and perform interpolation by ordinary kriging. Fig. 2 illustrates the difference between the unconditional covariance $R_{YY}(h)$ and the averaged conditional covariance $\bar{R}_c(h)$ associated with the kriging procedure. The averaged conditional covariance $\bar{R}_c(h)$ is almost isotropic and can be fitted by:

$$\bar{R}_c(h/\lambda) = 0.4932\sigma_Y^2 \exp\left(-\left(\frac{h}{0.435\lambda}\right)^{1.2056}\right) \tag{12}$$

For the given spacing of measurements $\Delta x = 1.25\lambda$, the conditional variance is about half as large as the unconditional variance, the integral scale is significantly smaller, and, at large separation distance, $\bar{R}_c(h)$ approaches zero more rapidly than $R_{YY}(h)$.

For each of both the fully resolved and the interpolated conductivity fields, the flow field is evaluated by the dual formulation (Frind and Matanga, 1985) yielding the distributions of hydraulic heads and stream function values. Assuming a mean head-gradient $\mathbf{J}=[-J,0]$ in the x_1-direction, the hydraulic heads ϕ in the infinite, periodic domain exhibits a linear trend with periodic fluctuations. Except for a constant, the field is

Table 2
Dimensionless parameters of the test case

Dimensions of domain	Spatial discretization
$\frac{L_1}{\lambda} = 50 \frac{L_2}{\lambda} = 25$	$\Delta \frac{x_1}{\lambda} = \Delta \frac{x_2}{\lambda} = 0.05$
Spacing of Measurements	
$\frac{\Delta x_{1,\mathrm{obs}}}{\lambda} = \frac{\Delta x_{2,\mathrm{obs}}}{\lambda} = 1.25$	
Covariance model isotropic exponential Local dispersion scalar	Variance of Y $\sigma_Y^2 = 0.25$ Péclet number $Pe = \frac{K_g J \lambda}{\theta D} = 500$
Damköhler Number	Inflow Concentration
$Da = \frac{\gamma \lambda^2}{D\sqrt{c_b^{\mathrm{in}} c_a^0}} = 500$	$\frac{c_b^{\mathrm{in}}}{c_a^0} = 1$

fully described by the hydraulic head distribution within a single unit cell with periodic boundary conditions (Kitanidis, 1992; Dykaar and Kitanidis, 1992):

$$\phi(L_1, x_2) = \phi(0, x_2) - JL_1 \tag{13}$$

$$\phi(x_1, L_2) = \phi(x_1, 0) \tag{14}$$

in which L_1 and L_2 are the length and the width of the unit cell. Because the fields of log-hydraulic conductivity $Y(\mathbf{x})$ and the hydraulic head fluctuations $\phi(\mathbf{x}) - \bar{\phi}(\mathbf{x})$ are periodic, the fields of specific discharge $\mathbf{q}(\mathbf{x})$ and seepage velocity $\mathbf{v}(\mathbf{x})$ are also periodic and thus fully described by the distribution within a single cell. This implies the following periodic boundary conditions for the stream function ψ:

$$\psi(L_1, x_2) = \psi(0, x_2) \tag{15}$$

$$\psi(x_1, L_2) = \psi(x_1, 0) + \bar{q} L_2 \tag{16}$$

in which $\bar{q}$ is the mean specific discharge evaluated from postprocessing of the head-calculation. From the ϕ- and ψ-fields, I generate unit cells bounded by streamlines and iso-potential lines using the streamline-oriented grid generator of Cirpka et al. (1999a). For illustration see Fig. 3. It may be noted that I use streamline-oriented grids in the

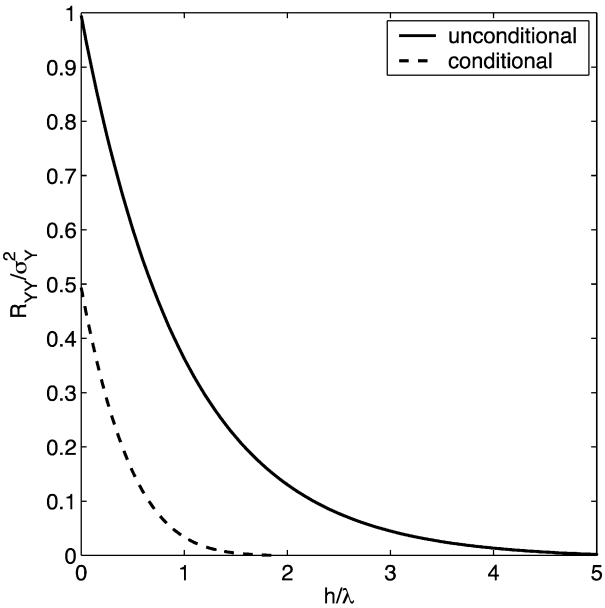

Fig. 2. Covariance of log-conductivity fluctuations. Solid line unconditional covariance $R_{YY}(h)$; dashed line averaged conditional covariance $\bar{R}_c(h)$ for $\Delta x_{obs}/\lambda = 1.25$.

subsequent transport calculations only to ensure that the simulations are not affected by numerical transverse dispersion (Cirpka et al., 1999b). The general approach to use the longitudinal correction effective dispersion coefficient $D^e_{c\ell}(t)$ for the approximation of

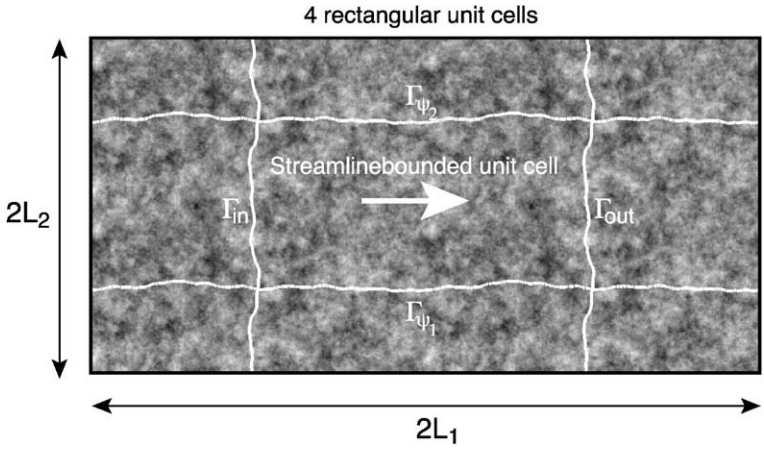

Fig. 3. Clip of a periodic log-conductivity field divided into unit cells bounded by streamlines and isopotential lines.

longitudinal mixing in interpolated fields is neither restricted to periodic domains nor to simulations on streamline-oriented grids.

3.2. Dispersion coefficients

Fig. 4 shows the local diffusion coefficient D, the effective longitudinal dispersion coefficient $D^{e}_{\ell}(t)$, and the longitudinal correction effective dispersion coefficient $D^{e}_{\ell c}(t)$ as function of travel time for the geostatistical parameters of the test case. $D^{e}_{\ell}(t)$ and $D+D^{e}_{\ell c}(t)$ were calculated by substituting $R_{YY}(h)$ and $\bar{R}_{c}(h)$ into Eqs. (3)–(6), respectively. $\bar{R}_{c}(h)$ and therefore $D^{e}_{\ell c}(t)$ do not depend on the actual values of the log-conductivity measurements. In the following reactive transport calculations, I apply D to simulations on fully resolved fields, $D+D^{e}_{\ell c}(t)$ to those on interpolated ones, and $D^{e}_{\ell}(t)$ in homogeneous domains. Already after passing a few integral scales, the values of $D^{e}_{\ell c}(t)$ are considerably larger than those of D, indicating the importance of small-scale variability on effective mixing in heterogeneous domains.

Fig. 5 shows the difference between longitudinal macrodispersion and effective dispersion coefficients. The travel time shown makes a plume sample one unit length of the periodic domain. The solid line applies when the effects of the full variability on transport is parameterized as dispersion. Within the given travel time, the difference between macrodispersion and effective dispersion is rather large because macrodispersion approaches its asymptotic value within a few integral scales while the catching up of effective dispersion is controlled by slow local transverse dispersion. The second increase of macrodispersion is an effect of periodicity. The dashed line in Fig. 5 shows the difference between longitudinal correction macrodispersion and correction effective dispersion coefficients. Obviously, the differences are much smaller. The correction dispersion coefficients refer to the unresolved small-scale variability. At these smaller scales, effective dispersion catches up with macrodispersion faster than at the scales resolved on the kriged fields.

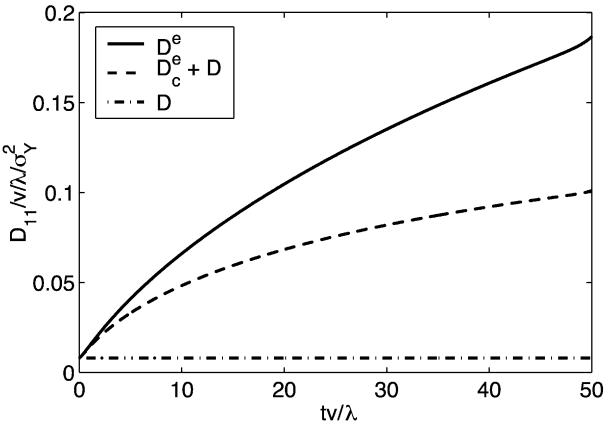

Fig. 4. Effective longitudinal dispersion coefficient D^{e}_{ℓ}, local diffusion coefficient D, and longitudinal correction effective dispersion coefficient $D^{e}_{c\ell}$ as function of travel time.

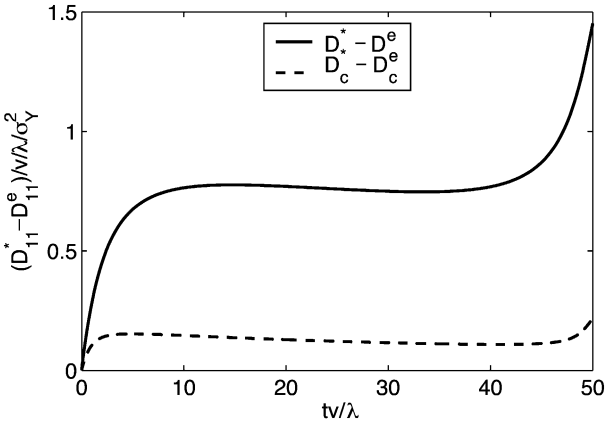

Fig. 5. Difference between longitudinal macrodispersion and effective dispersion as function of travel time.

3.3. Bimolecular reaction

As an example of mixing-controlled reactive transport, I simulate a second-order irreversible bimolecular reaction of non-sorbing compounds (Kapoor et al., 1997). I assume advective–dispersive transport of compounds A, B and C with concentrations c_a, c_b and c_c. Compounds A and B react with each other, forming compound C:

$$\mathrm{A} + \mathrm{B} \rightarrow \mathrm{C} \tag{17}$$

For simplicity, I assume all stoichiometric coefficients to be one. None of the compounds sorb. The governing partial differential equations are:

$$\frac{\partial c_a}{\partial t} = -\nabla \cdot (\mathbf{v}\mathbf{c}_a - \mathbf{D}\nabla c_a) - \gamma c_a c_b \tag{18}$$

$$\frac{\partial c_b}{\partial t} = -\nabla \cdot (\mathbf{v}c_b - \mathbf{D}\nabla c_b) - \gamma c_a c_b \tag{19}$$

$$\frac{\partial c_c}{\partial t} = -\nabla \cdot (\mathbf{v}c_c - \mathbf{D}\nabla c_c) + \gamma c_a c_b \tag{20}$$

in which γ is the kinetic coefficient of the bimolecular reaction. Its value is given in dimensionless form as Damköhler number *Da* in Table 2.

While the velocity field $\mathbf{v}(\mathbf{x})$ is periodic in all directions, the concentration fields are not. I consider a quasi one-dimensional replacement of a solution containing compound A by a solution containing compound B. For schematic illustration, see the top half of Fig. 1. Compound B is injected over an infinite cross section perpendicular to the direction of flow. Because of the periodicity in the velocity field, the concentration distributions are determined by a semi-infinite stripe having the width of a unit cell. In the following

simulations, I consider only the concentration distribution over the length of a single unit cell. Therefore, I define inflow- and outflow-conditions for a single unit cell in addition to the periodic conditions in the direction transverse to mean flow. For reasons of numerical accuracy, all calculations are performed on unit cells bounded by streamlines and iso-potential lines (see Fig. 3).

At the initial state, the concentration of compound A is uniformly distributed throughout the domain, whereas neither B nor C is present in the system:

$$c_{\mathrm{a}}(t_0,\mathbf{x}) = c_{\mathrm{a}}^0 \qquad c_{\mathrm{b}}(t_0,\mathbf{x}) = 0 \qquad c_{\mathrm{c}}(t_0,\mathbf{x}) = 0 \tag{21}$$

Compound B is introduced by advection into the domain over the inflow boundary Γ_{in}. For the outflow boundary Γ_{out}, the mass flux of all compounds is restricted to pure advection:

$$\begin{aligned}
\mathbf{n}\cdot(\mathbf{v}c_{\mathrm{a}} - \mathbf{D}\nabla c_{\mathrm{a}}) &= 0 && \text{on } \Gamma_{\mathrm{in}}\\
\mathbf{n}\cdot(\mathbf{v}c_{\mathrm{b}} - \mathbf{D}\nabla c_{\mathrm{b}}) &= \mathbf{n}\cdot\mathbf{v}c_{\mathrm{b}}^{\mathrm{in}} && \text{on } \Gamma_{\mathrm{in}}\\
\mathbf{n}\cdot(\mathbf{v}c_{\mathrm{c}} - \mathbf{D}\nabla c_{\mathrm{c}}) &= 0 && \text{on } \Gamma_{\mathrm{in}}\\
\mathbf{n}\cdot(\mathbf{D}\nabla c_{\mathrm{a}}) &= 0 && \text{on } \Gamma_{\mathrm{out}}\\
\mathbf{n}\cdot(\mathbf{D}\nabla c_{\mathrm{b}}) &= 0 && \text{on } \Gamma_{\mathrm{out}}\\
\mathbf{n}\cdot(\mathbf{D}\nabla c_{\mathrm{c}}) &= 0 && \text{on } \Gamma_{\mathrm{out}}
\end{aligned} \tag{22}$$

I apply periodic boundary conditions for the boundaries coinciding with streamlines:

$$\begin{aligned}
c_{\mathrm{a}} \mid \Gamma_{\psi_1} &= c_{\mathrm{a}} \mid \Gamma_{\psi_2} & \left.\frac{\partial c_{\mathrm{a}}}{\partial x_i}\right|_{\Gamma_{\psi_1}} &= \left.\frac{\partial c_{\mathrm{a}}}{\partial x_i}\right|_{\Gamma_{\psi_2}}\\
c_{\mathrm{b}} \mid \Gamma_{\psi_1} &= c_{\mathrm{b}} \mid \Gamma_{\psi_2} & \left.\frac{\partial c_{\mathrm{b}}}{\partial x_i}\right|_{\Gamma_{\psi_1}} &= \left.\frac{\partial c_{\mathrm{b}}}{\partial x_i}\right|_{\Gamma_{\psi_2}}\\
c_{\mathrm{c}} \mid \Gamma_{\psi_1} &= c_{\mathrm{c}} \mid \Gamma_{\psi_2} & \left.\frac{\partial c_{\mathrm{c}}}{\partial x_i}\right|_{\Gamma_{\psi_1}} &= \left.\frac{\partial c_{\mathrm{c}}}{\partial x_i}\right|_{\Gamma_{\psi_2}}
\end{aligned} \tag{23}$$

The periodic boundary condition facilitates dispersive exchange across the boundaries coinciding with streamlines. As explained above, I do not apply periodic boundary conditions to transport at the in- and outflow boundaries. Similar simulations have been performed by Cirpka and Kitanidis (2000a) who also accounted for sorption differences and related chromatographic mixing. Reactive transport is simulated by a cell-centered Finite Volume Method with a slope-limiter stabilization scheme for advection, an implicit time-integration scheme for dispersion, and an operator-split approach for the coupling of advective–dispersive transport and chemical reactions (Cirpka et al., 1999b). The change in concentrations due to the bimolecular reaction is calculated independently at each node for each timestep by an implicit Euler scheme.

For local dispersion, I apply D to simulations on the fully resolved fields, $D_\ell^e(t)$ to simulations in a homogeneous one-dimensional domain, whereas $D+D_{\ell c}^e(t)$ is used as longitudinal dispersion coefficient on the fields obtained from kriging. For comparison, I also perform simulations with D as local dispersion coefficient on the smoothed fields. In conservative-transport studies, transverse effective dispersion has been shown hardly affected by small-scale variability (Cirpka and Nowak, in press). Therefore, the transverse dispersion coefficient in the reactive-transport calculations on the kriged log-conductivity fields is assumed identical to the corresponding pore-scale dispersion coefficient. The correction effective dispersion coefficient depends on the travel time τ. Given the discharge Q_{st} of a stream tube and its cross-sectional area $A_{st}(\xi)$ as function of the distance ξ from the inlet, I calculate the travel time for each longitudinal interface of the streamline-oriented grid by:

$$\tau(x) = \frac{\theta}{Q_{st}} \int_0^x A_{st}(\xi) d\xi \tag{24}$$

and assign the corresponding dispersion coefficient to that interface.

3.4. Results

Fig. 6 shows the mean concentrations in the outflow of the fully resolved fields. The concentrations are averaged over all realizations. The breakthrough curves of compounds A and B show significant overlapping, which is contributed by the averaging procedure. At points within single realizations, compounds A and B hardly coexist because they react with each other (for illustration see, e.g., Fig. 2 of Cirpka and Kitanidis, 2000a). Certain parts of the outflow boundary experience an earlier breakthrough than others, transforming the sharp breakthrough curves of A and B to smoothed ones. The concentrations of product C are fairly small. The mean residence time within the domain being 50 times the

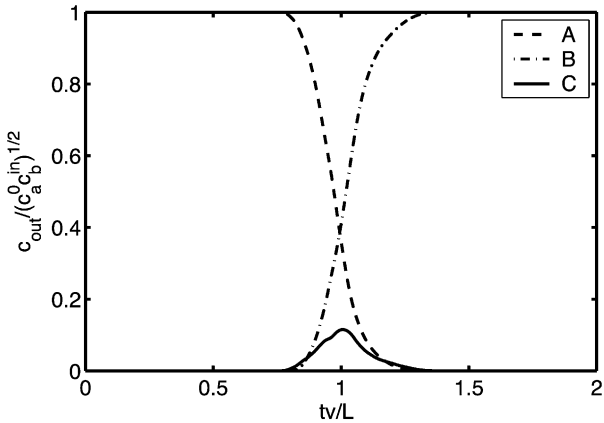

Fig. 6. Mean concentrations in the outflow of the domain. Calculations on the fully resolved fields.

characteristic time of reaction, the reaction is controlled by slow mixing rather than by reaction kinetics.

Fig. 7 shows the breakthrough curve of compound C in the outflow of the domain for two selected realizations. I compare the outflow concentrations on the fully resolved fields (denoted by "true") with those on the kriged fields using either D as local dispersion coefficient ("kriged") or $D+D^{e}_{c\ell}$ ("corrected"). In realization 1, the outflow concentrations on the fully resolved field and on the kriged fields using $D+D^{e}_{c\ell}$ as local dispersion coefficient are similar, whereas the simulations on the kriged fields using D as local dispersion coefficient underestimate the concentrations of compound C. In realization 9, by contrast to realization 1, the breakthrough curves denoted by "true" and "corrected" show rather large deviations. The different behavior of the breakthrough curves in the single realizations indicates that the unresolved small-scale features of the log-conductivity fields are very important in mixing at the travel distance considered.

Fig. 8 shows the ensemble average of the outflow concentration of compound C. The notation is identical to that of Fig. 7. In addition, Fig. 8 shows the breakthrough curve of

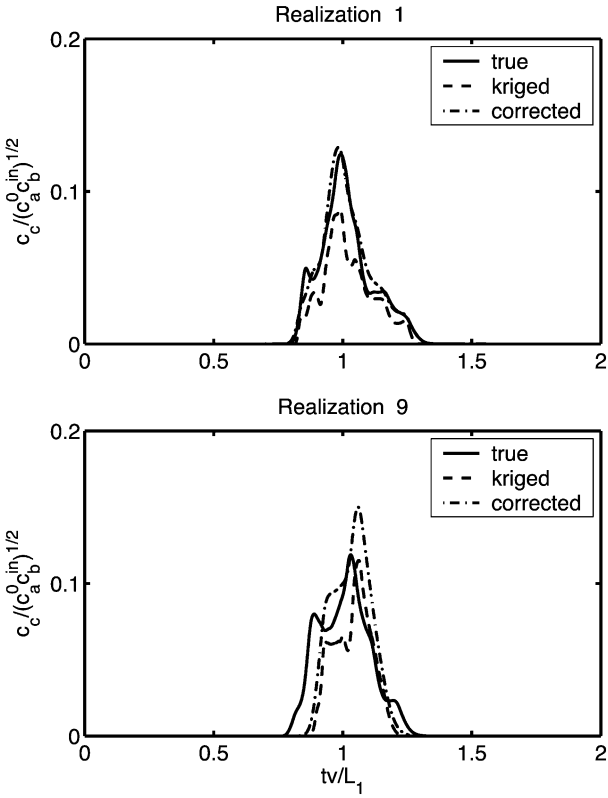

Fig. 7. Concentration of compound C in the outflow. Comparison between the fully resolved fields ("true"), the kriged fields using D as local dispersion tensor ("kriged"), and the kriged fields using $D+D^{e}_{c\ell}$ as local dispersion tensor ("corrected") for two selected realizations.

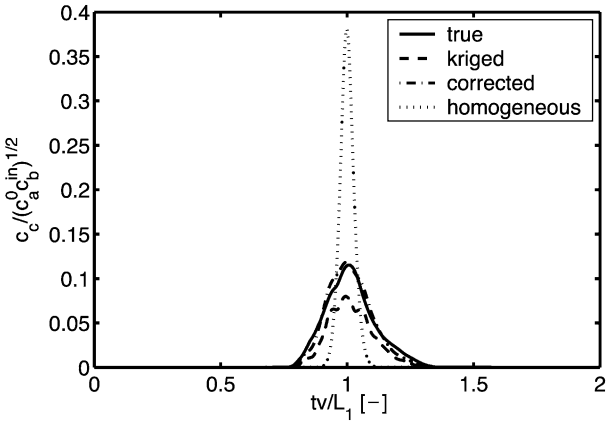

Fig. 8. Mean concentration of compound C in the outflow of the domain. Comparison between the fully resolved fields ("true"), the kriged fields using D ("kriged"), the kriged fields using $D+D^{e}_{c\ell}$ ("corrected"), and an equivalent uniform field using D^{e}_{ℓ} as local dispersion tensor ("homogeneous").

an equivalent homogeneous domain in which D^{e}_{ℓ} is used as local dispersion coefficient ("homogeneous"). In the ensemble average, the outflow concentrations on the fully resolved fields and on the kriged fields using $D+D^{e}_{c\ell}$ as local dispersion coefficient are even more similar than in realization 1 shown in Fig. 7. Again, the simulations on the kriged fields using D as local dispersion coefficient underestimate the concentrations of compound C. The simulations on the uniform field using the longitudinal effective dispersion coefficient D^{e}_{ℓ} show a breakthrough curve of compound C that is much more peaked than those from the fully resolved and kriged fields. However, the total mass of compound C leaving the domain appears to be similar. The results indicate that, in an average sense, applying the analytical expressions for the longitudinal (correction) effective dispersion coefficient D^{e}_{ℓ} and $D^{e}_{c\ell}$ is appropriate to guarantee the correct amount of mixing and, therefore, reaction rates in mixing-controlled reactive transport on equivalent uniform and smoothed log-hydraulic conductivity fields, respectively. For single realizations of the given problem, this need not to be valid, as indicated by realization 9 shown in Fig. 7. Note that the outflow concentrations shown in Fig. 7 are already averaged over the width of the domain. If the domain was wider, the cross-sectionally averaged concentrations would approach the ensemble average. In the present study, obviously, the domain is too narrow to average out the spatial variability in mixing and reaction rates.

Fig. 9 shows the ensemble-averaged mass of compound C in the domain as function of time. The mass is made dimensionless by:

$$m^{*}_{c} = \frac{1}{V}\int_{V} \frac{c_c(\mathbf{x})}{\sqrt{c_a^0 c_b^{in}}}\, d\mathbf{x} \tag{25}$$

As in the case of the breakthrough curves, we compare the simulated mass on the fully resolved fields (denoted by "true") with the mass on the kriged ones using either D as

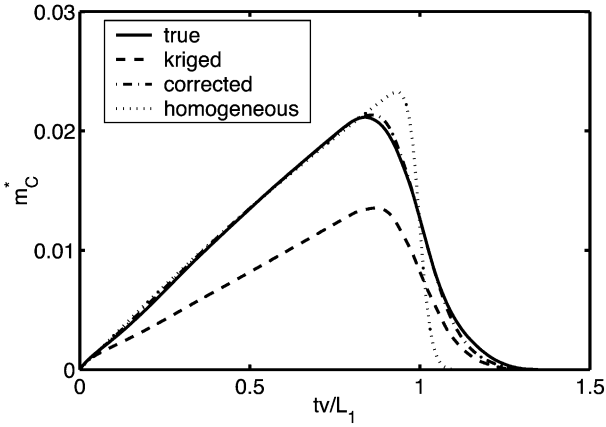

Fig. 9. Mean total mass of compound C in the domain. Comparison between the fully resolved fields ("true"), the kriged fields using D ("kriged"), the kriged fields using $D+D^{e}_{c\ell}$ ("corrected"), and an equivalent uniform field using D^{e}_{ℓ} as local dispersion coefficient ("homogeneous").

local dispersion coefficient ("kriged") or $D+D^{e}_{c\ell}$ ("corrected") and the mass on a equivalent uniform field using D^{e}_{ℓ} ("homogeneous"). The simulated masses on the fully resolved fields and the kriged fields using the corrected longitudinal dispersion coefficient as well as on the uniform field are almost indistinguishable, whereas the simulations on the smoothed fields without correction of the dispersion coefficient lead to a significant underestimation of the mass of compound C. When the plume leaves the domain, the simulated mass on the homogeneous field differs from the other calculations. This is contributed by the lacking spreading in the calculations on the uniform field.

In one-dimensional transport with constant coefficient of dispersive mixing and instantaneous reaction of compounds A and B, the mass of the reaction product C would increase by the square-root of time. Under these conditions, the production rate of C was controlled by the concentration gradients of A and B that decrease with time. In heterogeneous domains, the effective coefficient describing mixing increases with time, as shown in Fig. 4 for the test case. The combined effects of decreasing gradients and increasing effective dispersion coefficient lead at early times to a linear increase of mass C, which is observed in Fig. 9. From ≈ 0.85 pore volumes on, the mass of C decreases because the plume leaves the domain.

Fig. 10 shows the ensemble-averaged spreading $\sigma^2_{X_1}$ of compound C in the domain as function of time. The spreading is quantified by the second central spatial moment in the longitudinal direction normalized by the mass. The top half of Fig. 10 shows a comparison between the simulations on the fully resolved fields ("true") and on the kriged fields using $D+D^{e}_{c\ell}$ as local dispersion coefficient ("corrected") as well as on the equivalent uniform field using D^{e}_{ℓ} ("homogeneous"). The spreading increases over 40 integral scales. The subsequent decrease is caused by parts of the plume leaving the domain. The simulations on the fully resolved fields lead to a higher amount of spreading than on the interpolated fields. This behavior is anticipated since spreading is quantified by the difference between the macrodispersion coefficient and the effective dispersion coefficient. In the simulations

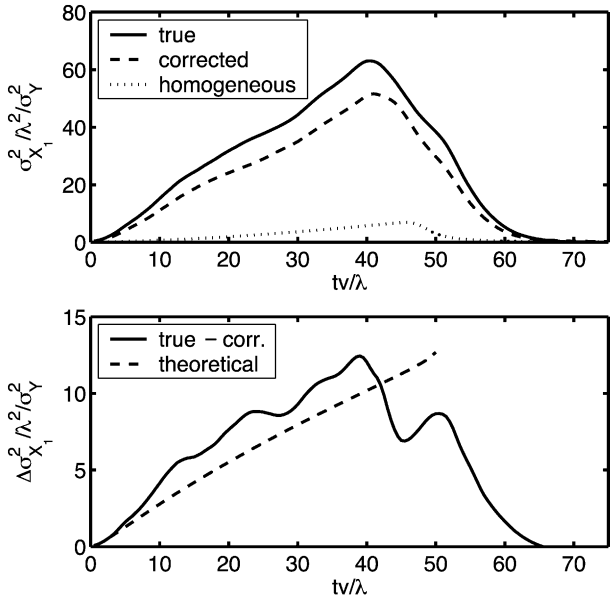

Fig. 10. Mean second central spatial moment of compound C. Top Comparison between calculations on the fully resolved fields ("true"), the kriged fields using $D+D^{\mathrm{e}}_{\mathrm{c}\ell}$ ("corrected"), and an equivalent uniform field using D^{e}_{ℓ} as local dispersion coefficient. Bottom Observed difference in the second central moment between the fully resolved and the kriged fields and theoretical prediction by $2\int_0^t (D^*_{c\ell}(\tau) - D^e_{c\ell}(\tau))d\tau$.

on the kriged fields, we correct only for the smoothing effects of kriging with respect to mixing but not spreading. The underestimation of spreading is much more severe for the calculations on the uniform field. The expected lack of spreading for the kriged field is quantified by $D^*_{\mathrm{c}\ell}(t) - D^{\mathrm{e}}_{\mathrm{c}\ell}(t)$, shown as dashed line in Fig. 5. Since the dispersion coefficient is half the rate of change of the second central moment, we can calculate the expected lack in ${\sigma_{X_1}}^2$ by:

$$\Delta\sigma^2_{X_1}(t) = 2\int_0^t (D^*_{\mathrm{c}\ell}(\tau) - D^{\mathrm{e}}_{\mathrm{c}\ell}(\tau))d\tau \tag{26}$$

This function is plotted as dashed line in the bottom half of Fig. 10. As long as the plume has not yet reached the outflow boundary, the predicted and observed lacks of spreading are in fairly good agreement. It should be noted, however, that Fig. 10 refers to the plume behavior averaged over all realizations. The agreement between prediction and simulations is not that good in single realizations.

4. Discussion and conclusions

The results of the present study strengthens the hypothesis that the effective dispersion coefficient of a conservative tracer introduced by a point source is an adequate measure of dispersive mixing in heterogeneous aquifers. Using this parameter in reactive-transport

simulations, I gained the correct overall rate of reaction. Since the effective dispersion coefficient is smaller than the corresponding macrodispersion coefficient of an extended plume, the simulations show a lack of spreading as shown in our simulations by the underestimation of the second central moment of the reaction product. The amount of lacking spreading can be approximated from the difference between longitudinal effective dispersion and macrodispersion coefficients. In a macroscopically uniform flow field, analytical expressions are available for both types of dispersion from linear stochastic theory (Fiori and Dagan, 2000; Dentz et al., 2000a).

When performing simulations on smoothed fields, only the effect of the unresolved variability needs to be parameterized as dispersion. The smoothing effect of kriging, or other geostatistical methods of conditioning, is quantified by the conditional covariance of the log-conductivity field. As the analytical expressions for effective dispersion and macrodispersion require stationarity, the nonstationarity of the conditional covariance must be removed by taking spatial averages. The resulting expressions are adequate for measurements on a regular, fairly dense grid. Substituting the averaged conditional covariance into the analytical expressions for effective dispersion, yields the coefficient required for reactive transport calculations on the kriged fields.

The methodology applies also to other smoothing techniques, provided a geostatistical characterization of the unresolved variability exists. In particular, it can be used in the context of geostatistical inverse methods in which the log-conductivity distribution is conditioned on hydraulic-head measurements (see, e.g., the methods compared by Zimmerman et al., 1998). Another important example would be the inference of the hydraulic conductivity distribution from geophysical data (Hubbard and Rubin, 2000). While the latter helps to identify larger structures in the subsurface associated with differences in the hydraulic conductivity, they cannot resolve small-scale features. Applying consistent conditioning techniques yields the geostatistical parameters needed to estimate mixing and dispersion caused by the unresolved heterogeneity.

In my simulations, the calculations on the fully resolved fields and on the interpolated ones using the corrected dispersion coefficient agree only in the ensemble average. Single realizations show higher deviations. These deviations indicate that mixing is sensitive to the details of small-scale heterogeneity. In field situations, these details cannot be identified. The cross-sectional average of plume properties approach the ensemble behavior if the plume is much wider than the integral scale of the heterogeneity (ergodic behavior). Therefore, the estimation of the mixing-related dispersion coefficients used in the present study leads to reliable results only if the ergodicity condition is met. The mass balance of narrow reactive plumes, by contrast, can be estimated only with a high uncertainty.

I consider smoothed fields. As a consequence of smoothing, the parameterization by dispersion can describe either mixing or macrodispersion, but not both. Similar results have been found by McLaughlin and Ruan (2001), who parameterized the unresolved heterogeneity by macrodispersion coefficients leading to correct second central moments of a conservative plume but wrong peak concentrations. The alternative to a single simulation on the smooth log-conductivity field of the best linear unbiased estimate would be to perform Monte-Carlo simulations using conditional realizations. Since conditional realizations exhibit all scales of heterogeneity, both mixing and spreading are approxi-

mated correctly. However, the computational effort of reactive-transport simulations is fairly high. In the present study, a single run requires approximately 1 day of CPU on a personal computer with a single 950-MHz processor. On this computer, Monte-Carlo simulations of reactive transport using several hundred realizations would not be feasible.

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